Case-disjoint benchmark of surrogate models for flow and thermal fields in supercritical CO_2 natural-circulation loops

Abstract

Surrogate models for supercritical carbon dioxide (sCO_2) natural-circulation loops (NCLs) can replace day-long steady-state conjugate computational fluid dynamics (CFD) simulations with millisecond-scale field predictions, yet most published evaluations rely on random node-level train–test splits or single pooled metrics that mask generalization failure across unseen operating conditions. Here we present a case-disjoint surrogate benchmark built from 42 steady-state RANS simulations solved with a k–ε family turbulence model (k and ε are exported) and a Span–Wagner equation of state verified against `CoolProp` (median 0.0008%) across three heater–cooler configurations (`HHHC`, `HHVC`, and `VHVC`) and two loop scales. Across three random seeds, we benchmark 17 surrogate models (plus an input-concatenation conditioning ablation) spanning tree ensembles, Fourier-coordinate neural networks, physics-informed neural networks (PINNs), DeepONet, Latent Neural Operator, Transolver-style attention, multi-scale graph networks, Fourier neural operators (`F-FNO`, `U-FNO`), and proper orthogonal decomposition (`POD`) reduced-order models under five-fold stratified case-disjoint cross-validation and leave-one-group-out extrapolation. We show that random node splitting understates five-fold case-disjoint error across the four nonlinear tree baselines by roughly an order of magnitude on thermodynamic state fields (6.78 × –10.37× on density, 11.29 × –87.54× on gauge pressure, and 5.62 × –10.24× on temperature in case-macro MAE) because high-capacity models memorize case-specific operating levels from adjacent mesh nodes. We also show that separating closed-loop gauge pressure into its case-level thermal-pressurization offset and centered spatial residual prevents small within-case hydrostatic variations from distorting model rankings. Across the eight exported CFD fields, no single architecture wins universally: among cross-configuration pooled models, `DeepONet` and `PINN continuity` lead on density and temperature, `Extra Trees` best predicts case-level gauge pressure, and multiplicative feature-modulated coordinates (`Domain-informed FiLM-MLP`) achieve the lowest error across both turbulence fields (k, ε) and all three velocity components (u, v, w), while within-group `POD-RBF` yields the sharpest temperature and turbulence reconstructions when interpolating on a fixed loop mesh.

Keywords: supercritical CO_2, natural-circulation loop, CFD surrogate benchmark, case-disjoint validation, neural operators, proper orthogonal decomposition

1 Introduction

Natural-circulation loops (NCLs) operating with supercritical fluids remove core and decay heat without mechanical pumps by balancing buoyancy forces between heated and cooled sections against frictional and local form losses [1, 2, 3]. Supercritical carbon dioxide (sCO_2, critical point $T_c = 304.13$ K $= 30.98°$C, $p_c = 7.3773$ MPa $= 73.773$ bar) is widely used as a scaled simulant fluid for supercritical water reactor (SCWR) thermal-hydraulics and compact power cycles because its lower critical pressure and temperature permit experiments and conjugate computational fluid dynamics (CFD) studies near the pseudocritical locus [4, 5, 6]. Near the pseudocritical temperature $T_{\mathrm{pc}}(p)$, small variations in enthalpy or wall heat flux induce steep nonlinear variations in density ρ, specific heat c_p, viscosity μ, and thermal conductivity k, coupling core-to-wall buoyancy gradients with turbulence production and heat-transfer deterioration [7, 8, 9].

Because resolving conjugate heat transfer between the primary sCO_2 pipe, the solid metallic piping, and the secondary cooling-water jacket requires fine near-wall meshes and real-gas property evaluations, each mesh-independent steady-state RANS simulation in our dataset requires approximately one day of compute time. Data-driven spatial surrogates—including gradient-boosted decision trees [10, 11, 12], physics-informed coordinate networks [13, 14, 15], operator-learning architectures [16, 17, 18, 19], and proper orthogonal decomposition (POD) reduced-order models [20, 21]—offer millisecond-scale inference once trained. However, three key methodological challenges limit current CFD surrogate evaluations for closed natural-circulation loops:

1. **Node-random data leakage:** Splitting a pooled table of millions of spatial nodes randomly between training and test sets places adjacent mesh nodes from every operating condition into both splits, reducing the task

to spatial interpolation within already-seen CFD solutions rather than generalization to unseen operating conditions.

2. **Metric distortion from closed-loop thermodynamic level offsets:** Reporting a single pooled R^2 across all test nodes conflates between-case operating-level shifts with within-case spatial profile accuracy. Conversely, in closed loops where thermal pressurization shifts the mean gauge pressure across cases by more than 1.7 MPa while the within-case spatial standard deviation is only ~ 1.5 kPa, naive within-case normalization produces uninterpretable pressure errors.

3. **Competing architectural inductive biases across thin-pipe loop geometries:** Closed circulation loops combine slender circular pipes ($D_i/L \sim 10^{-2}$), sharp 90° elbows, and distinct heater–cooler orientations, making it unclear when pointwise coordinate networks, operator-learning models, voxelized spectral convolutions, or mesh-aligned POD bases should be preferred across thermodynamic state versus kinematic and turbulence fields.

This paper addresses all three challenges through a three-seed surrogate benchmark across 42 steady-state sCO_2 natural-circulation loop CFD simulations. Specifically, our main contributions are:

- **Quantification of node-split leakage versus case-disjoint generalization:** Across four nonlinear tree ensembles (`Random Forest`, `Extra Trees`, `XGBoost`, `CatBoost`) and two low-capacity controls (`Mean baseline`, `Ridge`), we quantify the error inflation between 80/20 random node splits and five-fold case-disjoint cross-validation under both case-macro MAE and median case-normalized MAE (nMAE).

- **Thermodynamic audit and two-part decomposition of closed-loop gauge pressure:** We evaluate the empirical pressure–temperature slope (≈ 0.21–0.23 MPa K^{-1}) alongside the `CoolProp` Span–Wagner isochoric derivative $(\partial p/\partial T)_\rho = 0.315$–$0.365$ MPa K^{-1} and plane-weighted mean density variations across cases, characterize the near-stagnant steady solutions in `HHVC` at 20 W and 30 W, and introduce a pressure evaluation protocol that separates case-level mean error from centered spatial residual error.

- **Three-seed benchmark across 18 model variants and four structural groups:** We compare tree ensembles, Fourier-feature coordinate MLPs (isolating multiplicative FiLM modulation against additive input concatenation), physics-informed neural networks, DeepONet, Latent Neural Operator, Transolver-style attention, GAMS-GNN-style graph networks, factorized and U-Net Fourier neural operators (`F-FNO`, `U-FNO3D`), and within-group mesh-aligned methods (`POD-RBF`, `POD-GPR`, `Nearest case`) across three random seeds (`2026`, `2027`, `2028`), supplemented by four leave-one-group-out extrapolation studies.

2 Physical Domain and sCO_2 Natural-Circulation Loop Dataset

2.1 Loop Geometry, CFD Solver Configuration, and Primary Fluid Domain

The dataset comprises 42 converged steady-state RANS conjugate heat-transfer simulations of closed rectangular sCO_2 natural-circulation loops executed in ANSYS Fluent and exported on cross-sectional sampling planes via ANSYS CFD-Post. All 42 simulations use a k–ε family turbulence model (k and ε are exported) and a Span–Wagner equation of state verified against `CoolProp` [4, 22] (median relative density discrepancy 0.0008%) for the primary sCO_2 loop fluid. Mesh-independence checks were completed prior to parametric production runs, with each steady-state simulation requiring approximately 24 hours of compute time.

The 42 cases span three heater–cooler orientation configurations—horizontal heater and horizontal cooler (`HHHC`), horizontal heater and vertical cooler (`HHVC`), and vertical heater and vertical cooler (`VHVC`)—across two loop scales (Table 1): 29 small-loop cases ($L_x \times L_y = 0.8936 \times 1.0000$ m, inner pipe diameter $D_i = 15.00$ mm) and 13 large-loop `VHVC` cases ($L_x \times L_y = 3.0100 \times 4.0800$ m, $D_i = 13.84$ mm). Complete radial dimensions of the stainless-steel pipe wall and counterflow cooling-water jacket, axial heater/cooler coordinates, and cross-sectional sampling plane counts are provided in Supplementary Section S1 (Table S0a).

Each simulation is parameterized by four physical boundary/operating inputs encoded in its case identifier `{CONFIG}_P{Q}_{p}_bar_{mdot}kgs_T_{Tin}`: heater power $Q \in [10, 2500]$ W, primary-loop operating pressure $p_{\text{op}} \in \{80, 85, 90, 100\}$ bar, secondary cooling-water inlet mass flow rate $\dot{m} \in [0.010, 0.150]$ kg s^{-1}, and cooling-water inlet temperature $T_{\text{in}} \in \{283.5, 300, 305, 315\}$ K (Fig. 1, where the filename typo `HHHC_P2000_80_bar_0.05kgs_T_205` is mapped to its physical coolant inlet temperature $T_{\text{in}} = 305$ K, consistent with its minimum fluid temperature of 305.87 K). At each sampling node (x, y, z), eight CFD target fields are exported: density ρ (kg m^{-3}), gauge pressure p_{gauge} (Pa), temperature T (°C), turbulence eddy dissipation

ε ($m^2\ s^{-3}$), turbulence kinetic energy k ($m^2\ s^{-2}$), and Cartesian velocity components (u, v, w) ($m\ s^{-1}$). Excluding solid pipe/jacket wall nodes (`null` velocity/turbulence fields, $\rho \approx 7970$ kg m^{-3}) and outer cooling-water jacket nodes ($\rho > 950$ kg m^{-3}) and deduplicating intersecting plane coordinates isolates 5, 451, 341 unique primary sCO_2 pipe-core nodes ($\rho \in [153.2, 892.5]$ kg m^{-3}; Fig. 2).

Table 1: Summary of the 42 steady-state CFD cases by heater–cooler configuration and loop scale. Complete geometric and sampling-plane details appear in Supplementary Table S0a.

Config.	Scale	Cases	$L_x \times L_y$ (m)	D_i (mm)	Q (W)	p_{op} (bar)	$\dot{m}$ (kg s^{-1})	sCO_2 nodes/case
`HHHC`	Small	20	0.894×1.000	15.00	1000–2000	80–100	0.050–0.150	97,264
`HHVC`	Small	5	0.894×1.000	15.00	10–50	90	0.050	89,771
`VHVC`	Small	4	0.894×1.000	15.00	10–40	90	0.050	89,647
`VHVC`	Large	13	3.010×4.080	13.84	250–2500	85	0.010–0.035	207,586
Total	**4 groups**	**42**	–	–	**10–2500**	**80–100**	**0.010–0.150**	**5,451,341 total**

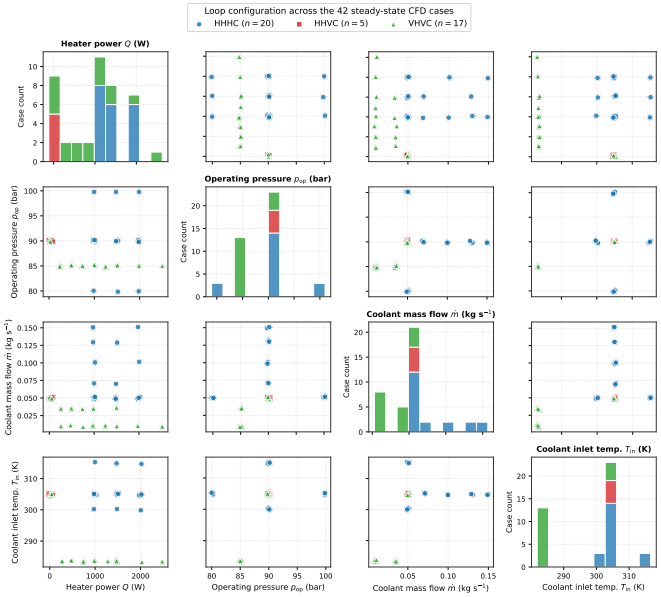


Figure 1: Operating-condition distribution across the 42 steady-state CFD cases by heater–cooler configuration (`HHHC`, `HHVC`, `VHVC`). Diagonal panels show stacked case-count histograms for heater power Q, operating pressure p_{op}, cooling-water inlet mass flow rate $\dot{m}$, and cooling-water inlet temperature T_{in} (with `T_205` mapped to 305 K).

2.2 Gauge Pressure, Empirical Thermal-Pressurization Slope, and Fixed-Mass Test

For thermophysical property evaluation across all 42 simulations, the local absolute pressure $p_{abs}(\mathbf{x})$ is related to the nominal operating pressure p_{op} and exported gauge pressure $p_{gauge}(\mathbf{x})$ by

$$p_{abs}(\mathbf{x}) = p_{op} + p_{gauge}(\mathbf{x}). \tag{1}$$

Evaluating the Span–Wagner CO_2 equation of state in `CoolProp` [4, 22] at $(T(\mathbf{x}), p_{abs}(\mathbf{x}))$ from Eq. (1) reproduces the exported nodal density $\rho(\mathbf{x})$ across all 42 cases to a median relative error of 0.0008% (case-level range 0.00004%–0.0015%), confirming that $p_{gauge}(\mathbf{x})$ enters the solver's equation-of-state lookup directly through Eq. (1). Within any single simulation c, the spatial variation of p_{gauge} around the loop is small because the solver subtracts a constant reference hydrostatic head (whose reference density ρ_0 is not exported in the CFD-Post tables, yielding a reduced vertical gradient $dp_{gauge}/dy \approx +0.37$ to $+0.94$ kPa m^{-1} along the vertical legs): the within-case standard deviation $\sigma_c(p_{gauge})$ is 0.15–0.39 kPa in the 1.00 m small loops and 1.45–8.79 kPa in the 4.08 m large loop, with a 42-case mean of $\bar{\sigma}_c = 1.51$ kPa.

By contrast, the case-mean gauge pressure $\bar{p}_c = \frac{1}{N_c}\sum_{i=1}^{N_c} p_{gauge,c,i}$ spans 1.42 kPa to 1.763 MPa across the 42 cases ($SD(\bar{p}_c) = 337.9$ kPa, or $224 \times \bar{\sigma}_c$; Fig. 3):

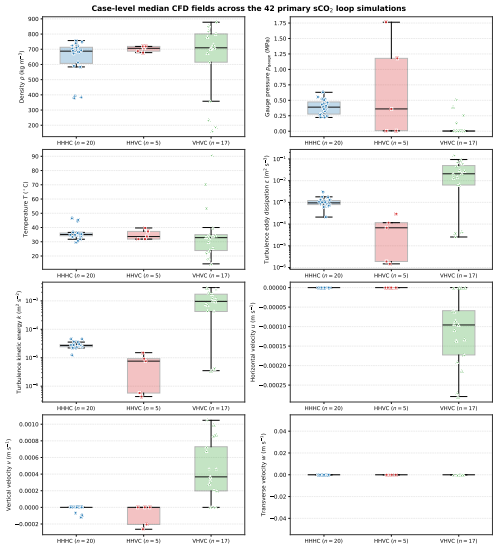


Figure 2: Case-level median distributions of the eight exported CFD fields across the 42 primary sCO_2 loop simulations ($5,451,341$ nodes), grouped by configuration (`HHHC`, `HHVC`, `VHVC`).

- **Top-pinned large loop versus unpinned small loops:** In the 13 large-loop `VHVC` cases, a gauge-pressure reference point is pinned near 0 Pa at the top of the loop ($y = 4.08$ m), keeping $\bar{p}_c \in [1.42, 8.77]$ kPa (Fig. 3a). In the 29 small-loop cases (`HHHC`, `HHVC`, and small `VHVC`), the closed sCO_2 domain has no pressure-outlet pin, so $\bar{p}_c$ rises with loop-mean temperature $\bar{T}_c$.

- **Empirical slope versus `CoolProp` $(\partial p/\partial T)_\rho$ and fixed-mass test:** Across the 17 small-loop cases at $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K ($\bar{\rho}_c \in [672.1, 718.0]$ kg m^{-3}, $\bar{T}_c \in [305.07, 312.51]$ K), ordinary least-squares regression of $\bar{p}_c$ against $\bar{T}_c$ yields an *empirical slope* of 0.214 MPa K^{-1} across all 17 cases and 0.235 MPa K^{-1} (≈ 0.23 MPa K^{-1}) across the five `HHVC` cases (Fig. 3b). Evaluating the theoretical Span–Wagner isochoric thermal pressure coefficient $(\partial p/\partial T)_\rho$ in `CoolProp` at each case's mean state $(\bar{\rho}_c, \bar{T}_c)$ gives 0.315–0.365 MPa K^{-1} (median 0.338 MPa K^{-1}, mean 0.341 MPa K^{-1}, and ≈ 0.35 MPa K^{-1} at $\rho = 710$ kg m^{-3}, $T = 305$–308 K; across all 29 small-loop cases, `CoolProp` $(\partial p/\partial T)_\rho$ spans 0.133–0.417 MPa K^{-1}, itemized in Supplementary Table S0b). To test whether total CO_2 mass is strictly fixed across cases within each configuration at identical $(p_{\mathrm{op}}, T_{\mathrm{in}})$, we compute the plane-weighted mean density $\bar{\rho}_{\mathrm{plane},c} = \frac{1}{P_c}\sum_{p=1}^{P_c} \bar{\rho}_{c,p}$ across all P_c cross-sectional planes. At $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K, $\bar{\rho}_{\mathrm{plane},c}$ is not strictly constant: it spans 671.6–704.0 kg m^{-3} (SD = 9.79 kg m^{-3}) across the eight `HHHC` cases, 681.2–717.8 kg m^{-3} (SD = 16.66 kg m^{-3}) across the five `HHVC` cases, and 696.8–712.1 kg m^{-3} (SD = 6.62 kg m^{-3}) across the four small `VHVC` cases (while in top-pinned `VHVC/large` at 85 bar, 283.5 K, $\bar{\rho}_{\mathrm{plane},c}$ drops from 860.8 to 161.5 kg m^{-3}, SD = 270.83 kg m^{-3}, as heating expands the fluid at near-constant pressure). Because $\bar{\rho}_{\mathrm{plane},c}$ decreases slightly as $\bar{T}_c$ increases across the small-loop cases ($\mathrm{d}\bar{\rho}/\mathrm{d}\bar{T} < 0$), the first-order thermodynamic relation $\mathrm{d}\bar{p} \approx (\partial p/\partial T)_\rho \,\mathrm{d}\bar{T} + (\partial p/\partial \rho)_T \,\mathrm{d}\bar{\rho}$ with $(\partial p/\partial \rho)_T > 0$ explains why the observed empirical slope (0.214–0.235 MPa K^{-1}) is lower than the pure isochoric `CoolProp` derivative (0.315–0.365 MPa K^{-1}).

- **Near-stagnant steady solutions in `HHVC` at 20 W and 30 W:** In the five `HHVC` cases at $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K (31.85°C), the non-monotonic variation of $\bar{p}_c$ with heater power $Q \in \{10, 20, 30, 40, 50\}$ W (Fig. 3a) arises because the $Q = 20$ W and 30 W runs (`HHVC_P20_90_bar_0.05kgs_T_305` and `HHVC_P30_90_bar_0.05kgs_T_305`) are near-stagnant steady solutions ($|\overline{\mathbf{u}}| = 0.9$–$1.1$ mm s^{-1}, $\bar{T}_c = 31.92$–31.94°C, $\bar{p}_c = 6.1$–9.2 kPa), whereas the $Q = 10$ W, 40 W, and 50 W runs have sustained buoyant circulation ($|\overline{\mathbf{u}}| = 0.044, 0.095, 0.107$ m s^{-1}, $\bar{T}_c = 33.58$°C, 37.09°C, 39.36°C, $\bar{p}_c = 0.361, 1.179, 1.763$ MPa) aligned with the empirical ≈ 0.23 MPa K^{-1} pressure–temperature trend (Fig. 3b).

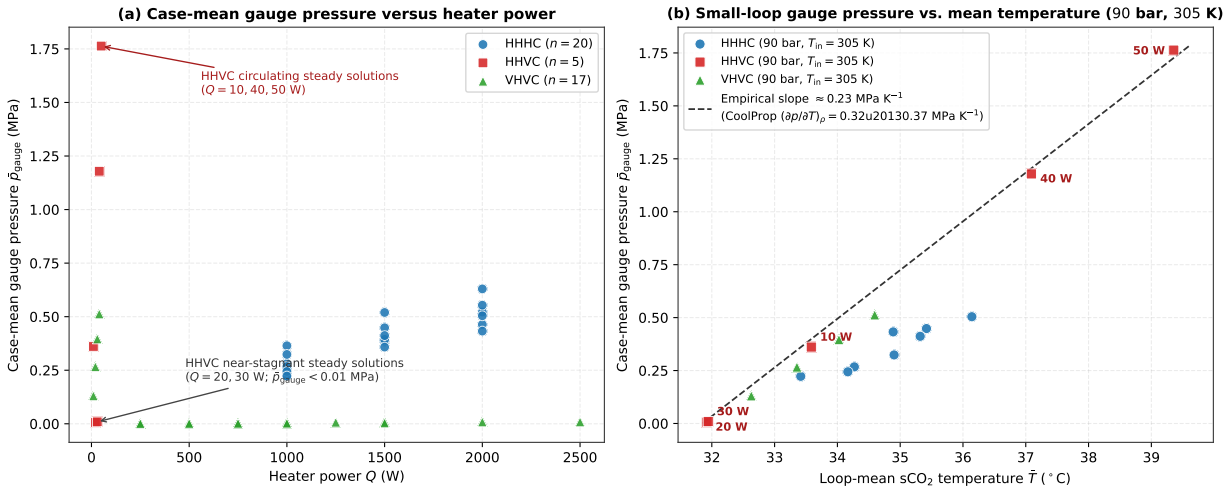


Figure 3: Case-level gauge-pressure behavior across the 42 sCO_2 simulations. (a) Case-mean gauge pressure $\bar{p}_{\mathrm{gauge}}$ versus heater power Q, contrasting top-pinned large-loop `VHVC` cases near 0 MPa with unpinned small-loop cases. (b) In the 17 small-loop cases at $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K, $\bar{p}_{\mathrm{gauge}}$ follows an empirical slope of 0.214 MPa K^{-1} (0.235 MPa K^{-1} across `HHVC`) versus loop-mean temperature $\bar{T}_c$, compared against the `CoolProp` Span–Wagner isochoric derivative $(\partial p/\partial T)_\rho = 0.315$–$0.365$ MPa K^{-1} (median 0.338 MPa K^{-1}), separating the near-stagnant steady solutions (20 W, 30 W) from the circulating solutions (10 W, 40 W, 50 W) in `HHVC`.

3 Benchmark Methodology, Models, and Evaluation Protocol

3.1 Validation Protocols and Evaluation Metrics

Every surrogate model predicts the eight exported field variables $\mathbf{y}(\mathbf{x}, \mathbf{c}) \in \mathbb{R}^8$ at query coordinates $\mathbf{x} = (x, y, z)$ given the case-level operating and geometric descriptor vector $\mathbf{c}$ (heater power Q, operating pressure p_{op}, coolant inlet mass flow rate $\dot{m}$, coolant inlet temperature T_{in}, loop extents L_x, L_y, pipe inner diameter D_i, and one-hot configuration indicators). We evaluate three splitting protocols:

1. **Five-fold stratified case-disjoint cross-validation (primary benchmark):** For each random seed $s \in \{2026, 2027, 2028\}$, the 42 cases are partitioned into five case-disjoint folds $(9, 9, 8, 8, 8$ cases) stratified by the four configuration/scale groups (`HHHC/small`, `HHVC/small`, `VHVC/small`, `VHVC/large`). Each seed independently re-randomizes the stratified five-fold case assignment, the four validation cases per training fold used for early stopping, the training node subsampling (up to 30,000 random nodes per training case in the three-seed GPU benchmark and 20,000 nodes per case in the CPU node-split versus case-split comparison), and neural weight initializations. All nodes of every held-out test case are unseen during training and standardization, and predictions are evaluated on 100% of the held-out case nodes (5,451,341 out-of-fold node predictions per seed).

2. **Within-group mesh-aligned cross-validation:** Mesh-aligned reduced-order models (`POD-RBF`, `POD-GPR`, and `Nearest case`) require a shared reference node ordering and therefore train and predict strictly within the matching configuration/scale group of each seed's five-fold split.

3. **Leave-one-configuration/geometry-group-out extrapolation:** All cases belonging to one of the four structural groups (`HHHC/small`, $n = 20$; `HHVC/small`, $n = 5$; `VHVC/small`, $n = 4$; or `VHVC/large`, $n = 13$) are held out simultaneously while the 16 cross-configuration pooled models train on the remaining three groups (16 models $\times$ 8 fields = 128 model–field evaluations per held-out group across three seeds).

For each held-out case $c \in \{1, \ldots, 42\}$ with N_c nodes and ground-truth values $y_{c,i}$ and predictions $\hat{y}_{c,i}$ for a given field, we compute the case-level MAE, RMSE, within-case standard deviation σ_c, and case-normalized MAE (nMAE$_c$):

$$\mathrm{MAE}_c = \frac{1}{N_c}\sum_{i=1}^{N_c} |\hat{y}_{c,i} - y_{c,i}|, \qquad \mathrm{RMSE}_c = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c} (\hat{y}_{c,i} - y_{c,i})^2}, \qquad \mathrm{nMAE}_c = \frac{\mathrm{MAE}_c}{\sigma_c}, \tag{2}$$

$$\sigma_c = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}(y_{c,i} - \bar{y}_c)^2}, \qquad R^2_{\text{pooled}} = 1 - \frac{\sum_{c=1}^{42}\sum_{i=1}^{N_c}(\hat{y}_{c,i} - y_{c,i})^2}{\sum_{c=1}^{42}\sum_{i=1}^{N_c}(y_{c,i} - \bar{y}_{\text{all}})^2}. \tag{3}$$

Across the 42 cases, we report the case-macro MAE ($\frac{1}{42}\sum_{c=1}^{42}\text{MAE}_c$), case-macro RMSE ($\frac{1}{42}\sum_{c=1}^{42}\text{RMSE}_c$), the median and interquartile range (IQR) of nMAE_c across cases, and the node-pooled R^2_{pooled} across all $N = \sum_c N_c = 5,451,341$ nodes.

Because $\bar{\sigma}_c(p_{\text{gauge}}) = 1.51$ kPa while case-mean gauge pressures span 1.763 MPa, dividing total node-level pressure error by σ_c conflates case-level thermal-pressurization offset errors with spatial gradient errors. We therefore suppress raw nMAE_c for p_{gauge} and decompose the gauge-pressure prediction into its case-mean level $\hat{\bar{p}}_c = \frac{1}{N_c}\sum_{i=1}^{N_c}\hat{p}_{c,i}$ and its centered spatial residual $\tilde{p}_{c,i} = p_{c,i} - \bar{p}_c$:

$$e_{\text{level},c} = |\hat{\bar{p}}_c - \bar{p}_c|, \quad \text{RMSE}_{\text{res},c} = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}\left(\hat{\tilde{p}}_{c,i} - \tilde{p}_{c,i}\right)^2}, \quad \text{nRMSE}_{\text{res},c} = \frac{\text{RMSE}_{\text{res},c}}{\max(\sigma_c, 1\text{ kPa})}, \tag{4}$$

where the 1 kPa denominator floor prevents division by near-zero spatial variance in the 1.00 m small loops ($\sigma_c = 0.15$–0.39 kPa).

3.2 Benchmark Model Suite and Verified Hyperparameters

Table 2 summarizes the 17 benchmark models (plus the `Domain-informed Concat-MLP` conditioning ablation and supplementary `POD-GPR`), and Table 3 reports their verified architecture and training hyperparameters extracted directly from the benchmark code (`notebooks/kaggle_clean_case_cv_predictions.ipynb` and `scripts/cpu_clean_benchmarks.py`).

To isolate multiplicative feature modulation in coordinate neural fields, `Domain-informed FiLM-MLP` and `Domain-informed Concat-MLP` share identical network depth (5 residual blocks), hidden width (256), $B = 4$ Fourier coordinate bands $\gamma(\mathbf{x}) = [\sin(2^b\pi\tilde{\mathbf{x}}), \cos(2^b\pi\tilde{\mathbf{x}})]_{b=0}^{3}$ [14] alongside pipe-leg centerline distance and arc-length features, and Huber loss ($\delta = 1.0$); `FiLM-MLP` modulates each hidden coordinate block via $\mathbf{h}^{(\ell)} \odot (1 + \boldsymbol{\gamma}^{(\ell)}(\mathbf{c})) + \boldsymbol{\beta}^{(\ell)}(\mathbf{c})$ [15], whereas `Concat-MLP` concatenates the operating descriptor $\mathbf{c}$ at the input layer. Both `PINN data-only` and `PINN continuity` use a 5-layer, width-256 Fourier-coordinate MLP with GELU activations and LayerNorm trained with MSE loss, where `PINN continuity` adds a penalty of weight $\lambda_{\text{cont}} = 0.05$ on the steady compressible continuity residual $\nabla \cdot (\hat{\rho}\hat{\mathbf{u}}) = \partial_x(\hat{\rho}\hat{u}) + \partial_y(\hat{\rho}\hat{v}) + \partial_z(\hat{\rho}\hat{w})$ evaluated by automatic differentiation on 1024 collocation nodes per step. Voxelized 3D neural operators (`F-FNO` and `U-FNO3D`) voxelize each case onto a $64 \times 64 \times 16$ Cartesian lattice with a binary occupancy mask and decode back to query mesh coordinates via trilinear grid sampling.

4 Results and Discussion

4.1 Optimism of Node-Random Splits Versus Case-Disjoint Validation

To quantify the data leakage induced by random node-level train–test splitting, we evaluate the four nonlinear tree ensembles (`Random Forest`, `Extra Trees`, `XGBoost`, `CatBoost`) alongside two low-capacity controls (`Mean baseline` and `Ridge`) under both an 80/20 random node split and five-fold case-disjoint cross-validation on the 5,451,341 primary sCO_2 nodes (seed `2026`). Table 4 and Figure 4 report the resulting error inflation ratios (case-disjoint error divided by random-node error).

As shown in Table 4 and Figure 4, the two low-capacity controls exhibit ratios near unity (`Mean baseline`: 0.95× –1.03×; `Ridge`: 1.00 × –1.26×) because a global constant or hyperplane cannot memorize local mesh neighborhoods. By contrast, across all four nonlinear tree models, moving from random-node splitting to five-fold case-disjoint validation inflates case-macro MAE by 11.29 × –87.54× on gauge pressure (11.29× for `CatBoost`, 13.79× for `Random Forest`, 26.15× for `Extra Trees`, and 87.54× for `XGBoost`), 6.78 × –10.37× on density, and 5.62 × –10.24× on temperature. Even on within-case normalized error, median case nMAE across the four nonlinear models increases by 3.34 × –4.05× on density and 2.95 × –4.39× on temperature, and by 1.34 × –1.61× on u and 1.10 × –1.29× on v. Under random node splitting, nonlinear models reconstruct a case's thermodynamic operating level by interpolating between adjacent nodes of the same simulation; under case-disjoint evaluation, the operating state must be inferred solely from $\mathbf{c}$.

Table 2: Summary of the evaluated surrogate model families and conditioning mechanisms.

Model	Family	Conditioning & spatial representation
`Mean baseline`	Baseline	Predicts the training-fold node mean for each target
`Ridge`	Linear	ℓ_2-penalized linear regression ($\alpha = 10.0$) on standardized $[\mathbf{x}, \mathbf{c}]$
`Random Forest`	Bagged trees	Multi-output CART ensemble on $[\mathbf{x}, \mathbf{c}]$ (48 trees in 3-seed CV; 80 in CPU split test)
`Extra Trees`	Randomized trees	Multi-output randomized trees on $[\mathbf{x}, \mathbf{c}]$ (64 trees in 3-seed CV; 96 in CPU split test)
`XGBoost`	Boosted trees	Histogram gradient-boosted trees fitted per target ('hist', depth 8)
`CatBoost`	Oblivious trees	Multi-output symmetric depth-8 gradient-boosted trees ('MultiRMSE')
`DeepONet`	Operator network	4-layer branch MLP on $\mathbf{c}$ + 4-layer trunk MLP on $\mathbf{x}$ (width 256, latent 128)
`PINN data-only`	Coordinate MLP	5-layer Fourier-coordinate MLP ($B = 4$, width 256) with pure data MSE loss
`PINN continuity`	Physics-informed	Identical to `PINN data-only` + $0.05\|\nabla \cdot (\hat{\rho}\hat{\mathbf{u}})\|_2^2$ on 1024 collocation nodes
`Domain-informed FiLM-MLP`	Modulated MLP	5-block Fourier + loop-feature residual MLP (width 256) with FiLM + Huber loss
`Domain-informed Concat-MLP`	Ablation MLP	Identical to `FiLM-MLP` but with input concatenation instead of FiLM modulation
`Latent Neural Operator`	Latent attention	Cross-attention from Fourier coordinates to $M = 24$ learned latent tokens (width 128)
`Transolver-style`	Physics attention	4-block slice-token attention across $S = 32$ learned physical slices (width 128)
`GAMS-GNN-style`	Graph network	4-block 3-scale graph message passing ($k = 6$ kNN + $20 \times 20 \times 6$ and $10 \times 10 \times 3$ voxels)
`F-FNO`	Spectral FNO	4-block factorized 1D spectral convolutions (width 24, modes 12) on $64 \times 64 \times 16$ grid
`U-FNO3D`	Multiscale FNO	3 spectral blocks coupled with a 3D U-Net branch on a $64 \times 64 \times 16$ grid
`POD-RBF` / `POD-GPR`	Aligned ROM	Within-group SVD modes (99.99% energy, ≤ 12 modes) + Gaussian RBF or GPR
`Nearest case`	Aligned lookup	Within-group nearest training case in standardized operating-parameter space

Table 3: Verified architecture and training hyperparameters from the benchmark execution scripts.

Model group	Verified hyperparameters in 3-seed GPU cross-validation (`kaggle_clean_case_cv_predictions.ipynb`)
Linear & tree ensembles (`Ridge`, `RF`, `ET`, `XGB`, `CB`)	$30,000$ subsampled training nodes/case ($20,000$ in CPU split test). `Ridge`: $\alpha = 10.0$. `RF`: 48 trees (80 CPU), max depth 20, max leaf nodes 4096 (2048 CPU), min leaf 5, max features 0.8, bootstrap max samples 0.3. `ET`: 64 trees (96 CPU), same leaf/depth settings. `XGB`: 300 trees (220 CPU), max depth 8 (6 on random-node CPU), lr 0.08, subsample 0.85, colsample 0.85, $\lambda = 2.0$. `CB`: 350 iterations, depth 8, lr 0.08, $\ell_2 = 3.0$, `MultiRMSE`.
`FiLM-MLP` & `Concat-MLP`	Width 256, 5 residual blocks, $B = 4$ Fourier bands, Huber loss ($\delta = 1.0$), AdamW (lr $= 5 \times 10^{-4}$, weight decay 10^{-5}), batch size 8192, sample budget 8×10^7 nodes (11 min wall-clock cap/fold), validation check every 5 epochs on 4 held-out training cases with patience 4.
`DeepONet` & `PINNs`	`DeepONet`: branch/trunk width 256, depth 4, latent dimension 128, MSE loss. `PINN data-only` / `PINN continuity`: width 256, depth 5, $B = 4$ Fourier bands, GELU + LayerNorm, MSE loss; `PINN continuity` adds $\lambda_{\text{cont}} = 0.05$ continuity residual on 1024 collocation points/step (9 min cap/fold).
`Latent Neural Operator`	Hidden width 128, $M = 24$ latent tokens, 4 attention heads, branch/query depth 3, decoder depth 2, $B = 4$ Fourier bands, Huber loss, AdamW (lr $= 5 \times 10^{-4}$), batch size 8192.
`Transolver-style` & `GAMS-GNN-style`	`Transolver-style`: width 128, 4 blocks, 4 heads, $S = 32$ slices, 8192 context / 4096 query nodes per case step, AdamW (lr $= 3 \times 10^{-4}$), 2200 steps (12 min cap). `GAMS-GNN-style`: width 96, 4 blocks, fine kNN ($k = 6$) on 8192 nodes + medium $(20, 20, 6)$ and coarse $(10, 10, 3)$ voxel graphs, AdamW (lr $= 2 \times 10^{-4}$), 1600 steps (14 min cap).
Voxel FNOs (`F-FNO`, `U-FNO3D`)	Grid $64 \times 64 \times 16$, spectral modes 12, width 24, 4 blocks (`F-FNO`) or 3 blocks + 3D U-Net (`U-FNO3D`), batch 2 cases, 4096 sampled points/case, AdamW (lr $= 10^{-3}$, weight decay 10^{-4}), 2000 steps (8 min cap).
Mesh-aligned (`POD-RBF`, `POD-GPR`, `Nearest case`)	Within-group SVD retaining 99.99% energy up to 12 modes. `POD-RBF`: leave-one-out CV over length scale $\ell \in \{0.25, 0.5, 1, 2, 4, 8\}$ and ridge $\alpha \in \{10^{-6}, 10^{-4}, 10^{-2}, 10^{-1}, 1\}$. `POD-GPR`: Gaussian kernel ($\ell = 1.0$, $\alpha = 10^{-4}$).

Table 4: Optimism of 80/20 random node splitting relative to five-fold case-disjoint cross-validation across the four nonlinear tree models (`Random Forest`, `Extra Trees`, `XGBoost`, `CatBoost`), with `Mean baseline` and `Ridge` shown separately as low-capacity controls. Ratios compare case-disjoint error to random-node error.

Model group & metric	Density ρ	Pressure p_{gauge}	Temp. T	TED ε	TKE k	Vel. u	Vel. v	Vel. w
4 nonlinear models: case-macro MAE ratio range	**6.78–10.37×**	**11.29–87.54×**	**5.62–10.24×**	**1.05–1.39×**	**1.41–2.99×**	**1.80–2.63×**	**1.57–2.34×**	**0.96–1.02×**
4 nonlinear models: median case nMAE ratio range	**3.34–4.05×**	n/a†	**2.95–4.39×**	**0.97–1.02×**	**0.99–1.11×**	**1.34–1.61×**	**1.10–1.29×**	**0.97–1.02×**
Control: `Mean baseline` MAE ratio (nMAE ratio)	0.96× (1.03×)	0.95× (n/a)	0.98× (0.96×)	1.00× (0.97×)	0.99× (1.02×)	1.00× (1.00×)	1.00× (1.00×)	1.00× (1.00×)
Control: `Ridge` MAE ratio (nMAE ratio)	1.26× (1.24×)	1.09× (n/a)	1.25× (1.26×)	1.01× (1.01×)	1.10× (1.02×)	1.00× (1.00×)	1.00× (1.00×)	1.00× (1.00×)

† Within-case nMAE is suppressed for gauge pressure because $\bar{\sigma}_c = 1.51$ kPa reflects only the within-case vertical pressure variation.

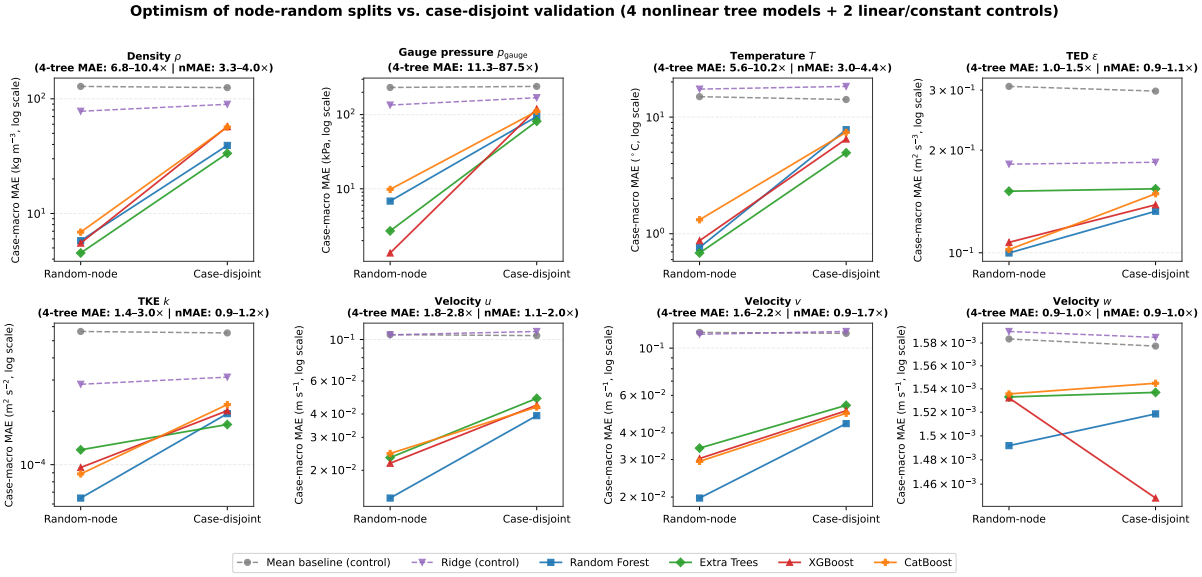


Figure 4: Case-macro MAE (log scale) under 80/20 random-node splits versus five-fold case-disjoint validation across the four nonlinear tree baselines (`RF`, `ET`, `XGB`, `CB`) and two controls (`Mean`, `Ridge`). Panel titles report the four-nonlinear-model range of error inflation under case-macro MAE and median case nMAE.

4.2 Three-Seed Case-Disjoint Benchmark Across All Eight CFD Fields

Figure 5 displays the three-seed (`2026`, `2027`, `2028`) case-macro MAE of all 18 primary models relative to the `Mean baseline` across the eight CFD fields. Table 5 reports the three-seed out-of-fold accuracy of the leading cross-configuration pooled model and leading within-group mesh-aligned model for each field, including the `Mean baseline` case-macro MAE, percentage MAE reduction relative to the `Mean baseline`, case-macro RMSE, median case nMAE (±SD across seeds), and pooled R^2 (full 19-model metrics appear in Supplementary Tables S1–S8).

Four main physical and architectural findings emerge from the three-seed benchmark:

1. **Thermodynamic state fields ($\rho, T, p_{\text{gauge}}$) versus kinematic and turbulence fields (u, v, w, k, ε):** Table 5 highlights why both case-macro MAE and within-case nMAE are needed. On density and temperature, pooled models reduce case-macro MAE by 73.3%–80.0% (ρ) and 60.0% (T) relative to the `Mean baseline` and achieve pooled R^2 of 0.862–0.936 and 0.736 by capturing the large between-case state shifts across $p_{\text{op}} \in [80, 100]$ bar and $Q \in [10, 2500]$ W. However, because the within-case standard deviation inside the small-loop cases is narrow ($\sigma_c(T) \approx 0.25$–2.5°C, $\sigma_c(\rho) \approx 5$–35 kg m^{-3}), even a modest case-mean offset yields a median case nMAE of 0.837 on ρ (`DeepONet`) and 1.350 on T (`PINN continuity`), which within-group `POD-RBF` lowers to 0.974 on T. Conversely, velocity and turbulence fields have near-zero case-mean offsets and strong geometry-locked spatial gradients; here `Domain-informed FiLM-MLP` reduces case-macro MAE by 78.4% on u, 80.6% on v, 73.9% on k, and 49.7% on ε, achieving median case nMAE of 0.162–0.242.

2. **Multiplicative FiLM modulation versus input concatenation:** Comparing `Domain-informed FiLM-MLP` against its matched ablation `Domain-informed Concat-MLP` (Fig. 5) shows that multiplicative FiLM conditioning reduces median case nMAE on six of the seven spatially varying fields: temperature (1.467 vs. 1.758), TED (0.192 vs. 0.226), TKE (0.242 vs. 0.273), u (0.162 vs. 0.224), v (0.175 vs. 0.230), and w (0.272 vs. 0.318), whereas additive input concatenation yields lower case-macro MAE on density (26.3 vs. 31.3 kg m^{-3}).

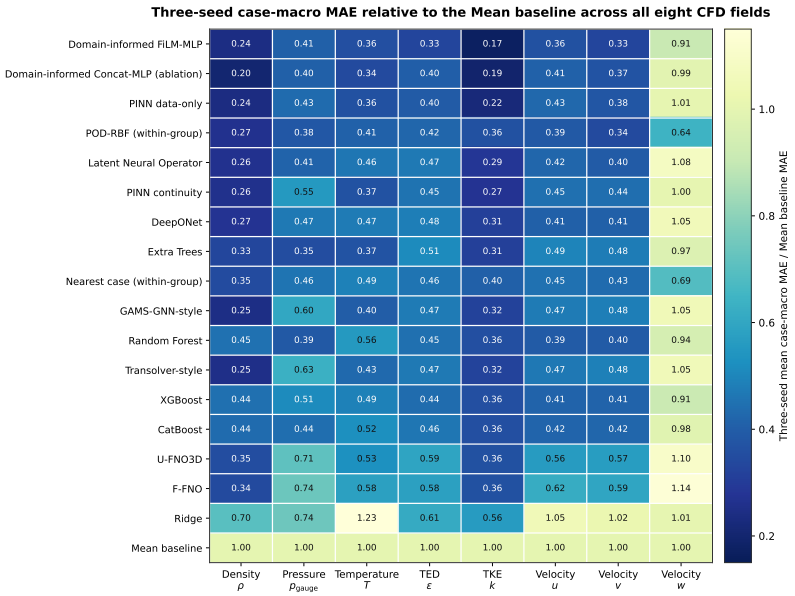


Figure 5: Three-seed mean case-macro MAE relative to the `Mean baseline` across all eight CFD fields for the 15 cross-configuration pooled models, the `Domain-informed Concat-MLP` conditioning ablation, and the two primary within-group mesh-aligned transfer models (`POD-RBF` and `Nearest case`), ordered by geometric-mean error ratio.

Table 5: Three-seed (`2026`, `2027`, `2028`) out-of-fold accuracy across the 42 primary sCO_2 cases. For each target, the first row gives the best cross-configuration pooled model (selected by median case nMAE for the seven spatially varying fields and by case-macro MAE for gauge pressure), followed by the best within-group mesh-aligned model where it achieves lower error, plus the lowest-MAE pooled model on density.

Target (unit)	Model	Scope	Macro MAE	Mean-base MAE	Reduction (%)	Macro RMSE	Median nMAE (±SD)	Pooled R^2
Density ρ (kg m^{-3})	`DeepONet`	Pooled 5-fold	35.1	131.6	73.3%	37.1	0.837 ± 0.160	0.862
	`Domain-informed Concat-MLP`	Pooled (lowest MAE)	26.3	131.6	80.0%	29.1	1.011 ± 0.069	0.936
Pressure p_{gauge} (kPa)	`Extra Trees`	Pooled 5-fold	83.2	254.0	67.3%	83.7	n/a† (2.91 res. nRMSE)	0.815
	`Nearest case` / `POD-RBF`	Within-group aligned	108.1 / 90.1	254.0	57.4% / 64.5%	108.2 / 90.2	0.0122 / 0.0483 res. nRMSE	0.776 / 0.571
Temperature T (°C)	`PINN continuity`	Pooled 5-fold	5.48	13.71	60.0%	7.32	1.350 ± 0.044	0.736
	`POD-RBF`	Within-group aligned	6.11	13.71	55.4%	7.42	0.974 ± 0.041	0.620
TED ε (m^2 s^{-3})	`Domain-informed FiLM-MLP`	Pooled 5-fold	0.0978	0.1945	49.7%	0.315	0.192 ± 0.020	0.569
	`POD-RBF`	Within-group aligned	0.127	0.1945	34.8%	0.369	0.177 ± 0.018	−0.575
TKE k (10^{-4} m^2 s^{-2})	`Domain-informed FiLM-MLP`	Pooled 5-fold	0.947	3.635	73.9%	1.57	0.242 ± 0.051	0.898
	`POD-RBF`	Within-group aligned	2.00	3.635	45.0%	2.78	0.237 ± 0.015	−0.070
Velocity u (m s^{-1})	`Domain-informed FiLM-MLP`	Pooled 5-fold	0.0381	0.1763	78.4%	0.0569	0.162 ± 0.015	0.801
Velocity v (m s^{-1})	`Domain-informed FiLM-MLP`	Pooled 5-fold	0.0383	0.1972	80.6%	0.0591	0.175 ± 0.004	0.861
Velocity w (10^{-3} m s^{-1})	`Domain-informed FiLM-MLP`	Pooled 5-fold	1.43	2.660	46.3%	3.46	0.272 ± 0.007	0.418
	`Nearest case` / `POD-RBF`	Within-group aligned	1.08 / 1.01	2.660	59.3% / 62.1%	2.93 / 2.55	0.185 ± 0.009 / 0.207 ± 0.008	0.551 / 0.650

† For gauge pressure, within-case nMAE is replaced by the centered spatial residual $\text{nRMSE}_{\text{res}}$ (1 kPa floor; Table S8).

3. **Gauge-pressure level versus spatial residual decomposition:** Total node-level gauge-pressure MAE (83.2 kPa for `Extra Trees`, a 67.3% reduction from the 254.0 kPa `Mean baseline`) is dominated by the case-level thermal-pressurization offset $e_{\text{level},c}$ (82.9 kPa for `Extra Trees`, driven by the contrast between the near-stagnant 20 W and 30 W `HHVC` cases near 8 kPa and the circulating 40 W and 50 W cases at 1.18–1.76 MPa). On the centered spatial residual $\tilde{p}_{c,i} = p_{c,i} - \bar{p}_c$ (Supplementary Table S8), mesh-aligned `Nearest case` ($\text{nRMSE}_{\text{res}} = 0.0122$), `POD-GPR` (0.0457), and `POD-RBF` (0.0483) and boosted trees (`XGBoost`, 0.151) preserve the smooth within-loop pressure profile, whereas unconstrained coordinate networks exhibit higher residual roughness ($\text{nRMSE}_{\text{res}} = 2.91$–$24.3$).

4. **Underperformance of voxelized 3D FNOs on slender pipe loops:** `F-FNO` and `U-FNO3D` rank near the bottom of the neural architectures across all fields (Fig. 5). Because the pipe inner diameter (13.84–15.0 mm) spans less than 1.7% of the 0.894×1.000 m and 3.010×4.080 m loop bounding boxes, voxelizing the loop onto a $64 \times 64 \times 16$ Cartesian grid places only one to two grid cells across the circular pipe cross-section, blurring radial boundary-layer gradients that pointwise coordinate networks and mesh-aligned `POD-RBF` resolve directly on the unstructured mesh.

Because the 42 simulations come from four structured parametric sweeps rather than an i.i.d. random sample,

we treat nonparametric Friedman repeated-measures tests and Holm-adjusted two-sided Wilcoxon signed-rank tests across the 42 cases (using three-seed mean per-case five-fold errors) as *exploratory* summaries (Supplementary Table S10). Across all eight fields, the Friedman test indicates overall model differences ($p < 0.001$). Under paired Wilcoxon signed-rank tests with Holm step-down adjustment ($\alpha = 0.05$) against `Domain-informed FiLM-MLP`:

- On turbulence fields (ε, k), `FiLM-MLP` achieves the lowest mean Friedman rank (4.62 on both ε and k). On ε, it is *significantly better than* all 15 pooled baselines (`Domain-informed Concat-MLP`, `PINN data-only`, `PINN continuity`, `DeepONet`, `Latent Neural Operator`, `Transolver-style`, `GAMS-GNN-style`, `F-FNO`, `U-FNO3D`, `Random Forest`, `Extra Trees`, `XGBoost`, `CatBoost`, `Ridge`, and `Mean baseline`; $p_{\text{Holm}} < 0.05$) and *not* significantly different from the three within-group mesh-aligned models (`POD-RBF`, `POD-GPR`, `Nearest case`; $p_{\text{Holm}} = 1.00$). On k, it is significantly better than all pooled baselines ($p_{\text{Holm}} < 0.05$) except its ablation `Domain-informed Concat-MLP` ($p_{\text{Holm}} = 0.190$), and not significantly different from the three within-group mesh-aligned models ($p_{\text{Holm}} \geq 0.190$).
- On horizontal and vertical velocities (u, v), `FiLM-MLP` achieves the lowest mean Friedman rank (5.83 on u, 5.26 on v). On v, it is *significantly better than* `Domain-informed Concat-MLP`, `PINN data-only`, `PINN continuity`, `DeepONet`, `Latent Neural Operator`, `Transolver-style`, `GAMS-GNN-style`, `F-FNO`, `U-FNO3D`, `Extra Trees`, `Ridge`, and `Mean baseline` ($p_{\text{Holm}} < 0.05$), and *not* significantly different from `CatBoost` ($p_{\text{Holm}} = 0.052$), `XGBoost` (0.157), `Random Forest` (0.286), or the within-group mesh-aligned models (`POD-RBF`, `POD-GPR`, `Nearest case`; $p_{\text{Holm}} \geq 0.641$). On u, it is significantly better than `Domain-informed Concat-MLP`, `PINN data-only`, `PINN continuity`, `Latent Neural Operator`, `GAMS-GNN-style`, `Transolver-style`, `F-FNO`, `U-FNO3D`, `Ridge`, and `Mean baseline` ($p_{\text{Holm}} < 0.05$), and not significantly different from `DeepONet` (0.550), the four tree ensembles ($p_{\text{Holm}} \geq 0.102$), or the within-group mesh-aligned models ($p_{\text{Holm}} = 1.00$).
- On transverse velocity w, `FiLM-MLP` (rank 7.02) is *significantly worse than* the three within-group mesh-aligned models (`POD-RBF`, `POD-GPR`, `Nearest case`; $p_{\text{Holm}} < 0.05$), *significantly better than* all other neural and operator baselines (`Domain-informed Concat-MLP`, `PINN data-only`, `Latent Neural Operator`, `DeepONet`, `GAMS-GNN-style`, `Transolver-style`, `F-FNO`, `U-FNO3D`; $p_{\text{Holm}} < 0.05$), and *not* significantly different from `PINN continuity`, the four tree ensembles, `Ridge`, or `Mean baseline` ($p_{\text{Holm}} \geq 0.096$).
- On density ρ and temperature T, `FiLM-MLP` is *significantly worse than* within-group `POD-RBF` on T ($p_{\text{Holm}} < 0.05$), *significantly better than* `Random Forest`, `XGBoost`, `CatBoost`, `POD-GPR`, `F-FNO`, `U-FNO3D`, `Ridge`, and `Mean baseline` on ρ and `Latent Neural Operator`, `F-FNO`, `U-FNO3D`, `Ridge`, and `Mean baseline` on T ($p_{\text{Holm}} < 0.05$), and *not* significantly different from `DeepONet`, `Domain-informed Concat-MLP`, `PINN continuity`, `PINN data-only`, `Transolver-style`, `GAMS-GNN-style`, or `Extra Trees` on either field ($p_{\text{Holm}} \geq 0.170$).

4.3 Spatial Reconstruction Analysis and Leave-One-Group-Out Extrapolation

Figure 6 displays held-out case-disjoint CFD reference fields, surrogate predictions, and absolute error maps $|\hat{y} - y|$ (seed `2026` out-of-fold predictions, annotated by nMAE_c rank among the 42 cases): `HHHC_P1500_90_bar_0.05kgs_T_300` for `DeepONet` on density ($\text{nMAE}_c = 0.517$, rank 15/42; 3-seed rank 3/42), `VHVC_P1000_85_bar_0.010kgs_T_283.5` for `PINN continuity` on temperature ($\text{nMAE}_c = 0.982$, rank 17/42; 3-seed rank 1/42), and `VHVC_P1000_85_bar_0.035kgs_T_283.5` for `Domain-informed FiLM-MLP` on vertical velocity v ($\text{nMAE}_c = 0.083$, rank 10/42; 3-seed rank 3/42). To provide an uncurated view across the full error distribution, Supplementary Figure S1 presents the corresponding spatial maps for the **median case (rank 21/42)**—`HHHC_P2000_100_bar_0.05kgs_T_305` on ρ ($\text{nMAE}_c = 0.636$), `HHHC_P1000_90_bar_0.07kgs_T_305` on T ($\text{nMAE}_c = 1.327$), and `HHHC_P1000_90_bar_0.13kgs_T_305` on v ($\text{nMAE}_c = 0.170$)—and the **worst case (rank 42/42)**—`HHHC_P2000_90_bar_0.05kgs_T_315` on ρ ($\text{nMAE}_c = 15.23$) and T ($\text{nMAE}_c = 16.00$), and `HHVC_P20_90_bar_0.05kgs_T_305` on v ($\text{nMAE}_c = 10.23$). As shown in Supplementary Figure S1, the median cases preserve the loop-scale density stratification and riser/downcomer velocity cores with localized errors in the cooler boundary layer and elbows, whereas the worst-case errors occur either at an extreme boundary condition (`T_315` at $Q = 2000$ W, where $\bar{T}_c = 47.5$°C exceeds the pseudocritical locus) or in the near-stagnant `HHVC` 20 W steady solution where $\sigma_c(v) = 1.5$ mm s^{-1}.

Table 6 summarizes the leave-one-configuration/geometry-group-out extrapolation study across all four structural groups (16 pooled models × 8 fields = 128 model–field pairs evaluated per held-out group across three seeds; full results in Supplementary Table S9). Holding out `VHVC/small` ($n = 4$) tests transfer across heater–cooler orientation when both the small-loop geometry (0.894 × 1.000 m, $p_{\text{op}} = 90$ bar, $T_{\text{in}} = 305$ K) and the vertical-heater/vertical-cooler orientation (via `VHVC/large`) are represented in training; here `XGBoost` and `Domain-informed FiLM-MLP` retain moderate median case nMAE on density (0.980), temperature (1.609), TED (0.268), TKE (0.391), and vertical

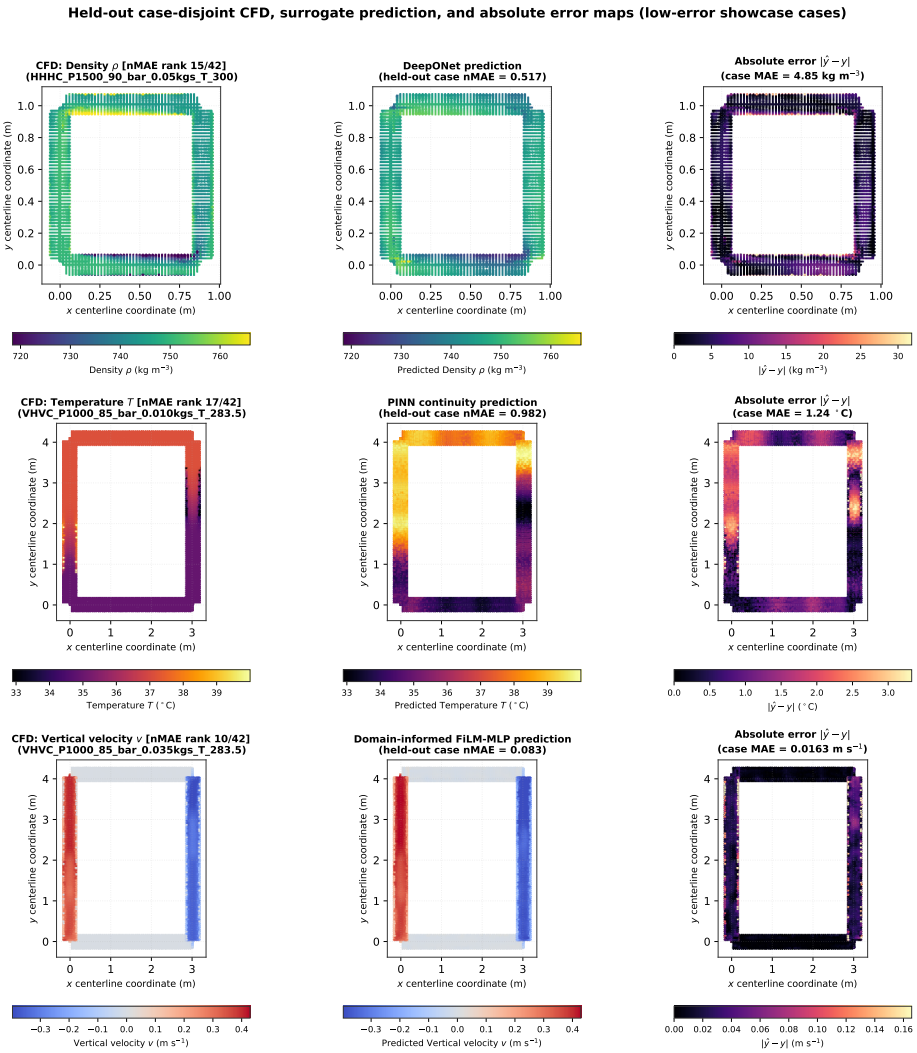


Figure 6: Held-out out-of-fold spatial field reconstructions (seed 2026) comparing reference CFD (left), surrogate prediction (middle), and absolute error $|\hat{y}-y|$ (right) for density ρ (`DeepONet`, case-nMAE rank 15/42), temperature T (`PINN continuity`, rank 17/42), and vertical velocity v (`Domain-informed FiLM-MLP`, rank 10/42). Corresponding median (rank 21/42) and worst (rank 42/42) cases for each field are shown in Supplementary Figure S1.

velocity v (0.686). By contrast, holding out `VHVC/large` ($n = 13$) forces simultaneous extrapolation across a 3.4 × –4.1× increase in loop dimensions (3.010×4.080 m), a different pipe diameter (13.84 mm), and an unseen coolant inlet temperature (283.5 K), raising the best density median nMAE to 3.854; similarly, holding out all 20 `HHHC/small` cases removes the only configuration with a top horizontal cooler and high-power small-loop operation (1000–2000 W), raising the best density and temperature median nMAE to 4.601 and 9.572.

Table 6: Leave-one-configuration/geometry-group-out extrapolation summary across the four structural groups. Each cell reports the best model and its three-seed mean median case nMAE among the 16 cross-configuration pooled models (16 models × 8 fields = 128 model–field pairs per held-out group; gauge pressure is reported as case-level MAE e_{level} and centered spatial residual $\text{nRMSE}_{\text{res}}$).

Field	Held-out `HHHC/small` ($n = 20$)	Held-out `HHVC/small` ($n = 5$)	Held-out `VHVC/small` ($n = 4$)	Held-out `VHVC/large` ($n = 13$)
Density ρ (nMAE)	`PINN cont.` (4.601)	`PINN data` (21.13)	`XGBoost` (0.980)	`Random Forest` (3.854)
Temp. T (nMAE)	`PINN cont.` (9.572)	`PINN data` (11.53)	`XGBoost` (1.609)	`PINN data` (1.682)
TED ε (nMAE)	`Transolver` (0.464)	`Concat-MLP` (0.822)	`FiLM-MLP` (0.268)	`GAMS-GNN` (0.258)
TKE k (nMAE)	`Concat-MLP` (1.198)	`Concat-MLP` (1.242)	`FiLM-MLP` (0.391)	`GAMS-GNN` (0.752)
Vel. u (nMAE)	`Extra Trees` (0.710)	`Mean base.` (0.869)	`Mean base.` (0.875)	`Concat-MLP` (0.810)
Vel. v (nMAE)	`XGBoost` (0.733)	`Mean base.` (0.873)	`XGBoost` (0.686)	`Concat-MLP` (0.766)
Vel. w (nMAE)	`Transolver` (0.687)	`Mean base.` (0.736)	`GAMS-GNN` (0.692)	`Mean base.` (0.558)
p_{gauge} level MAE (kPa)	`DeepONet` (254.0)	`Mean base.` (536.2)	`F-FNO` (226.4)	`Ridge` (204.2)
p_{gauge} res. $\text{nRMSE}_{\text{res}}$	`Mean base.` (0.273)	`Mean base.` (0.197)	`Mean base.` (0.172)	`Extra Trees` (0.734)

4.4 Limitations

Several limitations of the present dataset and benchmark should be noted:

1. **Unexported solver convergence histories and steady-branch ambiguity in `HHVC`:** Because only the final converged CFD-Post planar exports are available without solver residual histories or transient continuation logs, we cannot separate a physical flow-regime bifurcation from convergence to an alternative steady numerical branch in the two near-stagnant cases `HHVC_P20_90_bar_0.05kgs_T_305` ($Q = 20$ W) and `HHVC_P30_90_bar_0.05kgs_T_305` ($Q = 30$ W), nor can we verify the specific k–ε sub-variant, wall-function treatment, or unexported hydrostatic reference density ρ_0 beyond what is directly recoverable from the exported fields.

2. **Non-constant plane-weighted mean density across parametric cases:** As shown in Section 2.2 and Supplementary Table S0b, the plane-weighted mean CO_2 density $\bar{\rho}_{\mathrm{plane},c}$ varies across steady-state cases at the same $(p_{\mathrm{op}}, T_{\mathrm{in}})$ (SD = 6.62–16.66 kg m^{-3} in the small-loop groups), so the closed-loop cases represent independent steady solutions rather than a strictly constant-mass trajectory of a single sealed charge.

3. **Re-randomized folds across seeds and structured case coverage:** In our three-seed evaluation, seeds `2026`, `2027`, and `2028` re-randomize the five-fold stratified case partitions in addition to model initialization, so the across-seed standard deviations reflect combined partition and optimization variance. Moreover, the small sample sizes in `HHVC/small` ($n = 5$) and `VHVC/small` ($n = 4$) limit statistical power in group-holdout tests.

5 Conclusions

We have established a case-disjoint surrogate benchmark across 42 steady-state sCO_2 natural-circulation loop CFD simulations ($5,451,341$ primary sCO_2 nodes). Our main conclusions are:

1. **Random node splitting severely overstates surrogate accuracy and masks closed-loop level offsets:** Across the four nonlinear tree baselines (`Random Forest`, `Extra Trees`, `XGBoost`, `CatBoost`), 80/20 random node splits understate five-fold case-disjoint case-macro MAE by 6.78 × –10.37× on density, 11.29 × –87.54× on gauge pressure, and 5.62 × –10.24× on temperature (3.34 × –4.05× and 2.95 × –4.39× on median case nMAE for density and temperature), whereas low-capacity `Mean` and `Ridge` controls remain near 1.0 × –1.26×. Furthermore, because unpinned small loops exhibit large between-case thermal-pressurization offsets along an empirical slope of 0.214–0.235 MPa K^{-1} (`CoolProp` $(\partial p/\partial T)_\rho = 0.315$–$0.365$ MPa K^{-1}) while within-case pressure variations average only 1.51 kPa, decomposing gauge pressure into its case-level mean offset and centered spatial residual is essential for interpretable model evaluation.

2. **Complementary inductive biases of operator, FiLM-modulated coordinate, and POD models:** Across three random seeds under five-fold case-disjoint cross-validation, multiplicative feature modulation (`Domain-informed FiLM-MLP`) achieves the lowest error among pooled models on all three velocity components (nMAE = $0.162, 0.175, 0.272$) and both turbulence fields (nMAE = 0.192 on ε, 0.242 on k), outperforming additive input concatenation on six of seven spatially varying fields; `DeepONet` (nMAE = 0.837) and `PINN continuity` (nMAE = 1.350) lead among cross-configuration pooled models on density and temperature; and within a fixed loop mesh, `POD-RBF` achieves the lowest within-case nMAE on temperature (0.974), TED (0.177), and TKE (0.237). By contrast, uniform grid-splatted spectral operators (`F-FNO`, `U-FNO`) lose radial boundary-layer resolution due to the slender pipe aspect ratio ($D_i/L < 1.7\%$).

3. **Interpolation versus cross-scale geometric extrapolation:** Leave-one-group-out experiments across the four structural groups (128 model–field evaluations per group) show that pooled coordinate and tree models transfer across heater orientations when the loop scale and operating pressure are covered in training (holding out `VHVC/small`), but degrade when simultaneously extrapolating across a 3.4 × –4.1× change in loop dimensions, pipe diameter, and coolant inlet temperature (holding out `VHVC/large` or `HHHC/small`), motivating dimensionless arc-length and radial coordinate parameterizations for multi-scale loop surrogates.

Data and Code Availability

All preprocessing, CPU and GPU benchmark training, statistical testing, and figure/table generation scripts, along with the three-seed metric tables, case-level pressure/density audit CSV, and supplementary material (Section S1, Figure S1, and Tables S0a–S10), are publicly available at https://github.com/Barghav777/SCWR_3D.

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